

A Comprehensive Privacy-Preserving Federated Learning Scheme With Secure Authentication and Aggregation for Internet of Medical Things

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Abstract—Data mining, integration, and utilization are the inevitable trend of the Internet of Medical Things (IoMT) in the context of Big Data. With the increasing demand for data privacy, federated learning has emerged as a new paradigm, which enables distributed joint training of medical data sources without leaving the private domain. However, federated learning is suffering from security threats as the shared local model will reveal original datasets. Privacy leakage is even more fatal in healthcare because medical data contains critically sensitive information. In addition, open wireless channels are susceptible to malicious attacks. To further safeguard the privacy of IoMT, we propose a comprehensive privacy-preserving federated learning scheme with a tactful dropout handling mechanism. The proposed scheme leverages blind masking and certificateless proxy re-encryption (CL-PRE) for secure aggregation, ensuring the confidentiality of the local model and rendering the global model invisible to any parties other than clients. It also provides authentication of uploaded models while protecting identity privacy. Compared with other relevant schemes, our solution has better performance on functional features and efficiency, and is more applicable to IoMT systems with many devices.

Index Terms—Blind masking mechanism, CL-PRE, certificateless signature (CLS), federated learning, IoMT.

I. INTRODUCTION

WITH the rapid development of intelligent sensing and wireless communication, the Internet of Medical Things (IoMT) has become one of the most promising fields in smart healthcare. Distinguished from traditional healthcare, IoMT enables distributed smart medical devices (e.g., household medical equipment and mobile wearable devices) to collect real-time health data residing in different physical areas and transmit it to medical institutions for disease analysis [1], [2], [3]. Thanks to these IoMT devices, patients have the capacity to master their physical indicators without going to the hospital for unnecessary medical tests. Consequently, IoMT can lighten the burden on hospitals and provide patients with efficient and convenient healthcare services, such as chronic disease prevention [4] and telehealth diagnosis [5].

The above advanced medical services are indispensable to the assistance of machine learning. As a representative data analysis technology, machine learning can dramatically excavate the internal value of medical data to help doctors conduct more accurate diagnoses. Traditionally, smart healthcare relies on a central server to consolidate data for model training [6]. However, as medical data increases exponentially, the server suffers from a heavy storage and computation burden. Meanwhile, since IoMT systems involve vast sensitive physical information, direct sharing of raw medical data will inevitably result in severe data abuse and privacy leakage, and even pose threats to patients' health and lives [7], [8].

To cope with these problems, federated learning [9] emerges as a new paradigm, making it possible to train a united model on distributed datasets. On the one hand, federated learning takes pressure off the server by dispersing a complex learning task to edge nodes. On the other hand, instead of handing over complete training data, participants can simply transmit model parameters to attain a high-quality model [10]. Therefore, compared with the centralized solution, federated learning can not only mitigate privacy risk, but also improve communication

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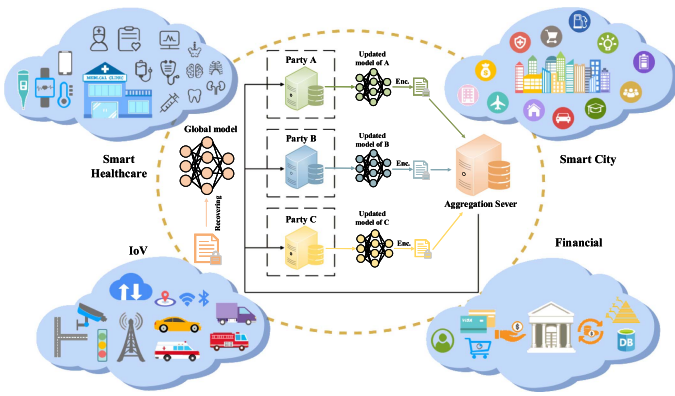


Fig. 1. Application scenarios of federated learning.

efficiency and scalability significantly. Based on these benefits, federated learning has been widely applied in many scenarios such as finance, IoMT (as shown in Fig. 1), changing the way of data circulation and creating a technical ecology of data available but not accessible in these fields.

Nevertheless, previous works [11], [12], [13], [14], [15] have pointed out that federated learning is still confronted with security and privacy issues. In essence, the shared model updates are calculated from the participant's private data, so attackers can intercept them and speculate relevant information in the original data samples. For example, some recent works train a binary classifier to determine if a particular instance (membership inference [11]) or an instance with a specific attribute (attribute inference [12], [13]) belongs to the training set. Furthermore, some attackers [14] develop generative adversarial network models to synthesize artificial samples similar to the training samples. Even worse, the original images and labels can be totally reconstructed through gradient monitoring [15].

In addition to gradient leakage, some new issues arise when applying federated learning in IoMT. First, since medical data is associated with patients' identities, it is imperative to protect identity privacy during collaborative training. Second, wearable medical devices may drop out at any time due to their rapid mobility and the unreliability of networks. Third, the openness of wireless channels makes it easy for malicious users to launch attacks like eavesdropping and tampering. To address these problems, we present a robust and privacy-preserving federated learning framework for IoMT. The main contributions of this work are summarized as follows:

- We introduce random masking and CL-PRE to design a lossless federated learning scheme. Among them, the blinding mask will be added to the local model, preventing attackers from speculating private samples. Moreover, CL-PRE allows the noise set by a third party to be securely shared with clients, and ensures it can be precisely eliminated in the global model.
- We design a secure client authentication mechanism based on traceable pseudonyms, which can protect identity privacy and resist man-in-the-middle (MITM) attacks during model uploading. Secure authentication is

needed to ensure that the sum of the noise is correctly removed.

- Considering the unreliable connection of some clients (such as mobile-based healthcare), we provide a fault-tolerant mechanism for federated learning. It can subtly eliminate the impact of dropped users, enable the normal proceeding of collaborative training, and will not incur unbearable communication overhead.

The remainder of this article is organized as follows. The related work of federated learning and certificateless cryptosystem are given in Section II. The preliminaries are introduced in Section III. The overview of the system model and threat model are discussed in Section IV. Section V describes the proposed scheme in detail. Section VI analyzes the correctness and security properties, and Section VII compares the performance of the proposed scheme with other relevant schemes. Finally, Section VIII concludes this article.

II. RELATED WORK

Federated learning is a distributed approach that allows collaborative training between multiple data sources. In federated learning, participants share local models rather than private samples. However, attackers can still infer the original data from model parameters. To further reduce privacy leakage, the existing privacy-preserving federated learning techniques mainly include three underlying routes: homomorphic encryption (HE), differential privacy (DP), and secure multi-party computation (SMPC).

HE enables direct arithmetical operations on the ciphertext without decryption. Up to now, a variety of federated learning schemes have adopted HE to protect model privacy. Zhang et al. [16] designed a cross-device federated learning framework FedSky for the mobile crowdsensing scenarios. FedSky first selects qualified training nodes by the size of local datasets and the computing capacity of mobile devices, then securely aggregates their model updates based on additive HE. Wibawa et al. [17] proposed a federated deep learning scheme for COVID-19 detection, which protects the privacy of local model through the fully homomorphic encryption algorithm BFV. Zhou et al. [18] employed the combination of blinding mask and Paillier to construct a secure federated learning scheme in fog computing, which not only protects the local gradients but also resists the collusion attack between the aggregation server and fog nodes. However, the above schemes are not suitable for training large-scale models because of the tremendous encryption and transmission overhead.

DP protects individual privacy by adding noise with specific distribution to the data. It requires the insertion or deletion of any single data will not significantly affect the outputs, thus concealing the sensitive personal information involved in the training set [19]. Jayaraman et al. [20] proposed a scheme to safeguard client-level privacy, in which the server adds Laplacian noise to the aggregated model updates, hiding the contribution of individual participants to the global model. However, the centralized method cannot resist attacks launched by malicious servers. In contrast, local differential privacy (LDP) is more feasible for

federated learning frameworks. Pihur et al. [21] utilized LDP for the spam classification federated task. In their scheme, the client uploads a perturbed model, preventing the server from spying on the actual local model parameters. Although the above DP-based schemes will not create excessive computation cost, the model accuracy is sacrificed to ensure an acceptable security requirement.

SMPC allows multiple parties to cooperatively evaluate a function without displaying private inputs. Kalapaaking et al. [22] proposed an SMPC-based federated learning scheme for IoMT, which implements secret sharing for secure aggregation and inference. Bonawitz et al. [23] constructed a secure aggregation protocol via double-masking and secret sharing, and it can sustain robustness in the case of network delay and user dropout. On this basis, So et al. [24] designed a multi-group circular mechanism for secure and fast model aggregation, in which the previous group passes masked results to the next for further aggregation. And Mandal et al. [25] improved [23] by introducing two non-colluding servers for non-interactive key generation. However, Their solutions still demand plentiful rounds of interaction for secret reconstruction, which incurs significant communication overhead.

Many of the above schemes leverage noise to perturb the local model, but the joining of noise poses a negative impact on model accuracy. To remedy this drawback, we introduce proxy re-encryption (PRE) to share the preset noise with clients and ensure the aggregated perturbations will be eliminated precisely. As an important cryptographic primitive, PRE empowers a third party as the proxy to convert the ciphertext, providing secure data sharing between the sender and receiver. At first, Blaze et al. [26] presented the primitive PRE scheme in the public key setting, which requires an authority to bind massive identities and certificates. To address the cumbersome certificate management problem, Green et al. [27] proposed the first IB-PRE scheme, but it incurs key escrow problem, i.e. the private key generator possesses the secret keys of all entities. Hence, Sur et al. [28] constructed a CL-PRE algorithm to avoid the aforementioned two drawbacks. Unfortunately, the scheme was not chosen ciphertext attack (CCA) secure as they claimed. Later, Guo et al. [29] designed a replayable CCA-secure CL-PRE scheme based on the random oracle model. Srinivasan et al. [30] proposed a pairing-free CL-PRE scheme to simplify the computation complexity.

To prevent the uploaded local models from being tampered with, we introduce certificateless signature (CLS) to verify the source and integrity of the models. Yum et al. [31] proposed the generic CLS scheme in 2004, but it was vulnerable to public key replacement attacks. After that, a series of researches [32], [33] were presented to improve its security. Recently, the CLS schemes [34], [35] have been used more frequently to guarantee identity privacy and message authentication.

III. PRELIMINARIES

In this section, we briefly introduce the background of federated learning and cryptographic preliminaries associated with our scheme.

A. Cryptographic Assumption

Definition 1 (Discrete Logarithm Problem (DLP)): Suppose $\mathbb{G}$ is a multiplicative cyclic group with order p , and g is a generator of $\mathbb{G}$. The DLP is that given $(g, g^a) \in \mathbb{G}$, there exists no probabilistic polynomial time (PPT) algorithm that can calculate $a \in \mathbb{Z}_p^*$ correctly, i.e., $\Pr[\mathcal{A}(g, g^a) = a] \leq \text{negl}(\cdot)$, where $\mathcal{A}$ is a PPT adversary and $\text{negl}(\cdot)$ is a negligible function.

Definition 2 (Computational Diffie-Hellman (CDH)): Let $\mathbb{G}$ be a multiplicative cyclic group of order p and g is a generator of $\mathbb{G}$. The CDH problem states that given $(g, g^a, g^b) \in \mathbb{G}$, where $a, b \in \mathbb{Z}_p^*$, the advantage of the PPT adversary to compute g^{ab} is negligible.

B. Horizontal Federated Learning

Horizontal federated learning is applicable to scenarios where the parties have the same feature space but different sample space [36]. A typical example is the multiple banks in disparate regions, whose customers may be completely different but the businesses are extremely similar. Given this circumstance, if these banks wish to co-train a model, they will perform the following five steps with the help of the aggregation server.

- *Step 1:* The server selects and initializes the target model.
- *Step 2:* All participants download the current global model, compute the gradients on their private datasets, and then mask the local model through some security methods such as HE, DP. Finally, they upload the masked model updates to the server.
- *Step 3:* The server aggregates these masked models without learning any information about the local model.
- *Step 4:* The server distributes the aggregation results to all participants.
- *Step 5:* The participants decrypt the results to acquire the latest global model, and update their local model respectively.

Repeat the last four steps until the pre-defined maximum iteration or training time is reached. In the end, the joint model will be shared among all participants.

C. Certificateless Proxy Re-Encryption

Certificateless proxy re-encryption is a fundamental cryptographic primitive for secure data sharing. In CL-PRE scheme, the semi-trusted proxy can convert an original ciphertext encrypted by the data owner into a ciphertext prepared for the data receiver.

Definition 3 (CL-PRE): The standard certificateless proxy re-encryption consists of the following algorithms:

- **Setup**(λ) $\rightarrow (pp, msk)$: Given the security parameter λ , the key generation center (KGC) generates the master secret key msk and the system parameter pp . Then KGC publishes pp and stores msk secretly.
- **KeyGen**(pp, msk, ID): it contains four sub-algorithms:
 - 1) **PartKeyExt**(pp, msk, ID) $\rightarrow (ppk_{ID}, psk_{ID})$: KGC inputs pp, msk and the user's identity ID , and outputs the partial key pair (ppk_{ID}, psk_{ID}) .

- 2) **SetSecretValue**(pp, ID) $\rightarrow s_{ID}$: The user inputs pp and ID , and outputs a secret value s_{ID} .
 - 3) **SetPrivateKey**(pp, psk_{ID}, s_{ID}) $\rightarrow SK_{ID}$: The user generates the full private key SK_{ID} through the partial private key psk_{ID} and the secret value s_{ID} .
 - 4) **SetPublicKey**(pp, ppk_{ID}, s_{ID}) $\rightarrow PK_{ID}$: The user creates the full public key PK_{ID} through the partial public key ppk_{ID} and the public key pk_{ID} computed from s_{ID} .
- **ReKeyGen**($pp, ID_A, ID_B, SK_A, PK_B$) $\rightarrow RK_{A \rightarrow B}$: The delegator A runs this algorithm to create the re-encryption key associated with the delegatee B . It inputs the identity ID_A, ID_B , the full secret key SK_A of the delegator, the full public key PK_B of the delegatee, and outputs the re-encryption key $RK_{A \rightarrow B}$.
 - **Encrypt**(pp, ID_A, PK_A, m) $\rightarrow C$: This is the original encryption algorithm executed by the delegator A . It takes ID_A, PK_A and a message m as input, and generates the level-1 ciphertext C .
 - **ReEncrypt**($pp, ID_A, ID_B, RK_{A \rightarrow B}, C$) $\rightarrow C'$: Given ID_A, ID_B , and $RK_{A \rightarrow B}$, the proxy transfers the level-1 ciphertext C into level-2 ciphertext C' .
 - **Decrypt1**(pp, ID_A, SK_A, C) $\rightarrow m$: The delegator A inputs ID_A and SK_A , and decrypts the original ciphertext C to get message m .
 - **Decrypt2**(pp, ID_B, SK_B, C') $\rightarrow m$: The delegatee B inputs ID_B and SK_B , and decrypts the ciphertext C' to get the shared message m .

D. Certificateless Signature

Definition 4 (CLS): In a certificateless signature scheme, the **Setup** and **KeyGen** algorithms are the same as CL-PRE, and the remaining two algorithms are as follows:

- **Sign**(pp, ID, SK_{ID}, m) $\rightarrow \delta$: The signer inputs ID, SK_{ID} and a message m , and outputs the signature δ .
- **Verification**($pp, m, \delta, ID, PK_{ID}$) $\rightarrow true/false$: The verifier inputs the received signature tuple (m, δ) , the signer's identity ID and public key PK_{ID} , to check if the signature is valid.

IV. SYSTEM OVERVIEW

In this section, we will introduce the system model, attack model, and design objectives of the scheme.

A. System Model

The system model of the proposed scheme is presented in Fig. 2, which consists of five entities: key generation center (KGC), trace authority (TRA), noise parameter server (NPS), aggregation server (AS), and clients.

- 1) **Trace authority**: TRA assists clients in generating pseudonyms to obscure their identities. It discloses the client's real identity if malicious behavior is detected. Considering the demand for identity tracing in each round, TRA should stay online all the time.

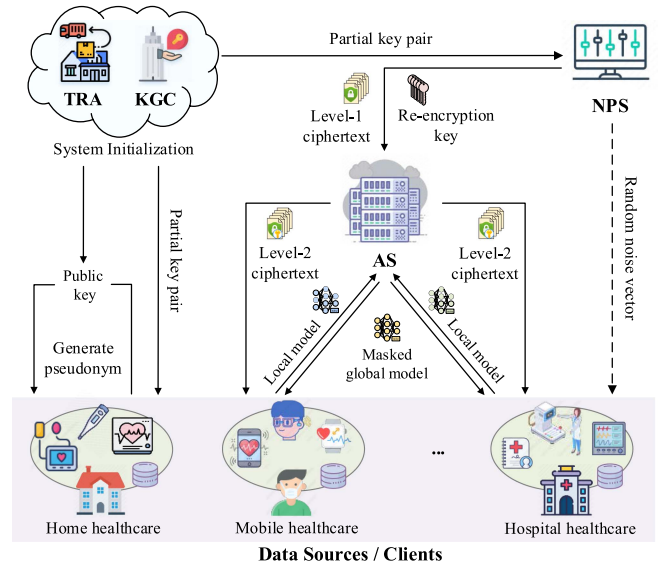


Fig. 2. System Overview.

- 2) **Key generation center**: KGC is mainly responsible for generating partial private keys for clients and NPS. It is worth noting that the partial keys of clients are linked to their pseudonyms rather than real identities.
- 3) **Noise parameter server**: NPS takes charge of creating blinding masks for clients' local models. It also serves as the delegator who intends to share these masks with clients through AS. Moreover, NPS primarily carries out pre-computation tasks and thus can be offline during the federated training phase.
- 4) **Clients**: Clients stand for mobile IoMT devices, medical institutions, and other healthcare facilities. They collect patients' physical information to constitute a database and conduct local training coordinated by AS. To avoid their privacy from being violated, clients add noise to the original models before uploading them to AS.
- 5) **Aggregation server**: AS functions as the proxy between NPS and clients. It re-encrypts the level-1 ciphertexts of random noise vectors and sends them to corresponding clients. When clients fulfill the local training, AS gathers their masked model parameters, securely aggregates them, and broadcasts the latest global model to clients.

B. Threat Model

The threat model is generally provided to analyze the potential security threats, which mainly includes two parts:

External threat: an external adversary $\mathcal{A}$ may have two ways to undermine the system security.

- 1) $\mathcal{A}$ could filch the local model parameters, aggregation results, and other private data by eavesdropping on the communication channels.
- 2) $\mathcal{A}$ might intercept and modify the client's model updates, then transmit the counterfeit messages to the AS, hindering the correct aggregation of the system.

Internal threat: the internal threat might involve five entities.

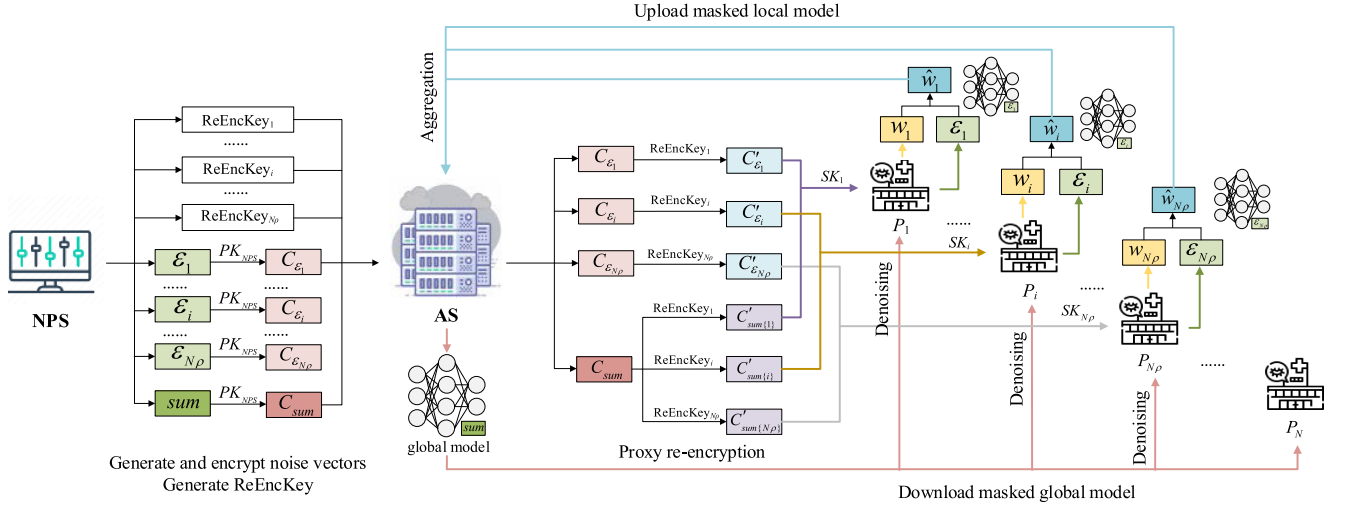


Fig. 3. Process of proxy re-encryption.

- 1) *TRA*: TRA is fully trusted and will not be compromised.
- 2) *KGC*: KGC is not completely credible. Generally, KGC is considered as an adversary who possesses the master private key, but cannot substitute user's public key.
- 3) *NPS*: NPS is always reliable for generating trustworthy blinding masks and it does not engage in the transmission of sensitive model parameters. Additionally, it is further assumed that NPS will not collude with AS.
- 4) *Clients*: The clients are assumed to be semi-honest. They will loyally conceal their local model with the allocated noise, but they are curious about others' private data.
- 5) *AS*: AS is an honest-but-curious auxiliary party. Although AS re-encrypts and aggregates honestly, it will speculate the training data from the received local updates, or try to acquire the final model for extra benefits.

C. Design Objectives

Our scheme aims to the secure aggregation of local models under the horizontal federated learning framework of IoMT. Considering the above security threats, we have the following design goals.

- 1) *Privacy Preservation*: The scheme is supposed to ensure the privacy of each client. To be precise, both identity privacy and data privacy need to be guaranteed. Furthermore, the global model should not be revealed to third parties other than the clients.
- 2) *Message Authentication*: To defend against malicious attacks during transmission, the message sent from clients needs some incidental information for authentication. Besides, AS ought to verify the legitimacy of a message and detect if it has been modified or falsified.
- 3) *Traceability*: To prevent pseudonyms from being abused by external adversaries, TRA should be able to analyze the controversial messages provided by AS and trace the real identity of the sender.

- 4) *Robustness*: Due to the instability of IoMT devices, the system is desired to be robust. Even if some clients in the distributed network disconnect from AS, the model aggregation should not be affected.

V. PROPOSED SCHEME

In this section, we first give an overview of the proposed secure and robust federated learning scheme for IoMT, then expound on each step in detail.

A. Overview

The privacy-preserving federated learning scheme involves four phases: system initialization, entity registration, pre-computation and secure aggregation. Among them, the proxy re-encryption of noise vectors plays a crucial role in privacy protection and lossless model training. Therefore, we present the elaborate process of PRE in Fig. 3. In addition, the main notations and descriptions are summarized in Table I.

B. System Initialization

Given the security parameter λ , TRA and KGC first choose a large prime p and a multiplicative cyclic group $\mathbb{G}$ with order q , where $\mathbb{G}$ is a subgroup of $\mathbb{Z}_p^*$. Let g be a generator of $\mathbb{G}$. Next, KGC randomly selects $x \in \mathbb{Z}_q^*$ as master private key, and computes the system public key $y = g^x$. TRA takes a random number $\mu \in \mathbb{Z}_q^*$ as the private key for identity tracing and computes $K_{pub} = g^\mu$. Then, TRA and KGC choose seven secure hash functions:

$$\begin{aligned}
 H_0 &: \mathbb{G} \times \{0, 1\}^* \rightarrow \mathbb{Z}_q^*, \\
 H_1 &: \mathbb{G} \times \mathbb{G} \times \{0, 1\}^* \times \{0, 1\}^* \rightarrow \mathbb{Z}_q^*, \\
 H_2 &: \{0, 1\}^{l_0} \times \{0, 1\}^{l_1} \times \{0, 1\}^* \times \mathbb{G} \rightarrow \mathbb{Z}_q^*, \\
 H_3 &: \{0, 1\}^* \times \mathbb{G} \times \{0, 1\}^* \times \{0, 1\}^* \times \mathbb{G} \rightarrow \mathbb{Z}_q^*, \\
 H_4 &: \mathbb{G} \rightarrow \{0, 1\}^l, \\
 H_5 &: \mathbb{G} \times \{0, 1\}^l \times \mathbb{G} \rightarrow \mathbb{Z}_q^*,
 \end{aligned}$$

TABLE I
NOTATIONS AND DESCRIPTIONS

Notation	Description
x, μ	Master secret key, trace key
P_i	Clients participated in federated learning
PID_i	The pseudonym associated with ID_i
TS_i	The timestamp of pseudonym PID_i
$(D, K), k$	Partial public key and partial private key
z	Secret value set by P_i or NPS
PK, SK	Full public key and full private key
$RK_{A \rightarrow B}$	Re-encryption key
S	A subset of clients selected for training
$\varepsilon_{t,i}$	A random noise vector for P_i in round t
sum_t	The sum of noise vectors in round t
C	A level-1 ciphertext generated by NPS
C'	A level-2 ciphertext transformed by AS
$w_t^{(i)}$	Veritable local model of P_i
$\hat{w}_t^{(i)}$	Concealed local update
δ_i	The signature of $w_t^{(i)}$
$\bar{w}_t$	Concealed global update
$\bar{w}_t$	Veritable global model
d	The number of dropped clients

$$H_6 : \mathbb{G}^4 \times \{0, 1\}^{l_0} \rightarrow \mathbb{Z}_q^*,$$

Here $l_0, l_1 \in N$, and $l = l_0 + l_1 \in N$. Finally they publish the system parameters $pp = (p, q, g, y, K_{pub}, H_i (i = 0, \dots, 6))$.

C. Entity Registration

- 1) *Pseudonym generation*: Each client $P_i (i \in [1, N])$ with real identity ID_i selects $t_i \in \mathbb{Z}_q^*$, and calculates $PID_{i,1} = g^{t_i}, PID_{i,2} = ID_i \oplus H_0((K_{pub})^{t_i} || TS_i)$, where TS_i denotes the timestamp of each training round. Then P_i sends the pseudonym $PID_i = \{PID_{i,1}, PID_{i,2}, TS_i\}$ to KGC.
- 2) *Partial Key Generation*: KGC chooses $s_i \in \mathbb{Z}_q^*$, computes $D_i = g^{s_i}, k_i = s_i + xH_1(D_i || PID_i), K_i = g^{k_i}$, sets $ppk_i = (D_i, K_i)$ as partial public key and $psk_i = k_i$ as partial private key for the client with pseudonym PID_i . Similarly, KGC generates $D_\xi = g^{s_\xi}, k_\xi = s_\xi + xH_0(D_\xi || ID_\xi), K_\xi = g^{k_\xi}$ for NPS, then transfers psk and ppk to the corresponding parties.
- 3) *User Key Generation*: After receiving the partial key pair, P_i picks out a secret value $z_i \in \mathbb{Z}_q^*$, computes $Q_i = g^{z_i}$, and the full public-private key pair is $SK_i = (k_i, z_i), PK_i = (D_i, K_i, Q_i)$. NPS also generates the public-private key pair (SK_ξ, PK_ξ) in the same way, which will be used for subsequent re-encryption key generation and original encryption, respectively.

Before generating the re-encryption key for client authorization, NPS first verifies the validity of public keys by checking

$$K_i \stackrel{?}{=} D_i y^{H_1(D_i || PID_i)} \quad (1)$$

to ensure the binding relationship between the public key and the homologous pseudonym.

- 4) *Re-encryption Key Generation*: To securely share the noise vectors with clients, NPS calculates

$$u_i = H_3(ID_\xi || PID_i || (Q_i K_i)^{z_\xi})$$

which links NPS with the authorized client. Then the re-encryption key is

$$RK_{NPS \rightarrow P_i} = (k_\xi + z_\xi) u_i^{-1}.$$

D. Pre-Computation

- 1) *Noise Vector Generation*: Let ρ be the percentage of clients participating in each round of training, and the selected clients constitute S , $|S| = N\rho$. NPS generates random noise vectors $\varepsilon_{t,i}$ for clients, where t represents the number of rounds and i represents that the vector is generated for P_i . The length of $\varepsilon_{t,i}$ should be relatively large to provide a sufficient level of masking. Note that in horizontal federated learning, the parameter dimensions are exactly the same for all local models, so the dimension of $\varepsilon_{t,i}$ should be concordant with them.

Throughout the training task, NPS needs to generate $T \cdot N \cdot \rho$ random vectors $\{\varepsilon_{t,i}\}_{i=1}^{N\rho} (t = 1, \dots, T)$, and calculate $sum_t = \sum_{i=1}^{N\rho} \varepsilon_{t,i}$ of each round.

- 2) *Encryption*: In this phase, NPS will encrypt $\varepsilon_{t,i}$ and sum_t with PK_ξ , respectively. To be more specific, firstly, NPS encodes and maps the vector into a message $m \in \{0, 1\}^{l_0}$, fills it with $\sigma \in \{0, 1\}^{l_1}$, and computes $\alpha = H_2(m || \sigma || ID_\xi || Q_\xi)$. Next, NPS chooses a random number $\beta \in \mathbb{Z}_q^*$ and acquires level-1 ciphertext $C = (C_1, C_2, C_3, C_4)$, where

$$\begin{aligned} C_1 &= g^\alpha, C_3 = g^\beta \\ C_2 &= H_4((Q_\xi \cdot K_\xi)^\beta) \oplus (m || \sigma) \\ C_4 &= \alpha + \beta H_5(C_1 || C_2 || C_3) \end{aligned}$$

Finally, NPS sends pseudonyms and re-encryption keys $\{PID_i, RK_{NPS \rightarrow P_i}\}_{i=1}^N$, together the ciphertexts $\{\{C_{\varepsilon_{t,i}}\}_{i=1}^{N\rho}, C_{sum_t}\}_{t=1}^T$ to AS. Once the above procedures are completed, NPS could be offline.

- 3) *Proxy Re-encryption*: Upon receiving the above tuples, AS firstly executes $CTIntegrityCheck(C)$ on each set of ciphertexts, which is given by

$$g^{C_4} \stackrel{?}{=} C_1 (C_3)^{H_5(C_1 || C_2 || C_3)} \quad (2)$$

If the verification passes, it indicates that the ciphertext has not been tampered with. Then AS stores all ciphertexts and re-encryption keys.

In each round, AS randomly allocates the training task to $N\rho$ clients, and selects $RK_{NPS \rightarrow P_i}$ to convert the ciphertext sent by NPS. AS computes $C'_1 = C_1, C'_2 = C_2, C'_3 = (C_3)^{RK_{NPS \rightarrow P_i}}$ and obtains level-2 ciphertext $C' = (C'_1, C'_2, C'_3)$ that could be decrypted by the corresponding client P_i .

Note that the original ciphertext AS needs to transform includes $C_{\varepsilon_{t,i}}$ and C_{sum_t} . The re-encryption ciphertexts of $C_{\varepsilon_{t,i}}$ will only be distributed to clients in S , while the process result of C_{sum_t} will be dispatched to N clients, to ensure that all clients can remove the masking of the global model in denoising phase.

- 4) *Noise Vector Decryption*: Clients in S receive two types of ciphertext, $C'_{\varepsilon_{t,i}}$ and C'_{sum_t} . As for the clients who are not included in S , they do not participate in this round of training and thereby have no need for the mask. However, considering that they might be selected in next round, it

Algorithm 1: Local Training.

Input: batch size s , max local iterations $MaxIter$, local dataset $\mathcal{D}_i$, learning rate η , latest real global model $\bar{w}_t$

Output: local model update $w_{t+1}^{(i)}$

Initialize $w_{1,1}^{(i)}$ to $\bar{w}_t$

for each local iteration $k = 1, 2, \dots, MaxIter$ **do**

Batches $\leftarrow$ randomly split $\mathcal{D}_i$ into multiple batches of size s , total batches per iteration $B = |\mathcal{D}_i|/s$

Set $w_{1,k}^{(i)}$ to the parameter of previous iteration $w_{1,k}^{(i)}$

for each batch index $b = 1, 2, \dots, B$ **do**

Compute the batch gradient $g_b^{(i)}$

Update the model parameter

$$w_{b+1,k}^{(i)} \leftarrow w_{b,k}^{(i)} - \eta g_b^{(i)}$$

end for

end for

Update the local model of this round

$$w_{t+1}^{(i)} = w_{B,MaxIter}^{(i)}$$

is necessary for them to get C'_{sum_t} . All of them decrypt the re-encryption ciphertexts to get the shared vectors. Specifically speaking, P_i computes $m \parallel \sigma = C'_2 \oplus H_4(U)$ where $u'_i = H_3(ID_\xi \parallel PID_i \parallel (Q_\xi)^{z_i+k_i})$ and $U = (C'_3)^{u'_i}$. Then, P_i checks the decryption result by

$$C'_1 \stackrel{?}{=} g^{H_2(m \parallel \sigma \parallel ID_\xi \parallel Q_\xi)} \quad (3)$$

If the equation holds, P_i maps m back to $\varepsilon_{t,i}$ or sum_t . After the above pre-computation, the noise vectors are distributed to clients. And the selected clients will conduct the subsequent training and masking.

E. Secure Aggregation

- 1) *Local training and Noise adding:* Client P_i in S runs Algorithm 1 to gain the local model $w_t^{(i)}$. For fear of model privacy leakage, the client utilizes noise to perturb the local model. After noise addition, $w_t^{(i)}$ turns to the concealed model parameter $\hat{w}_t^{(i)} = w_t^{(i)} + \frac{1}{n_{t,i}} \varepsilon_{t,i}$.
- 2) *Sign:* To sign a masked model $\hat{w}_t^{(i)}$, P_i encodes and maps it into plaintext space as $w \in \{0, 1\}^{l_0}$, which is the same as the first step of noise vector encrypting. Then P_i selects $r_i \in \mathbb{Z}_q^*$, calculates $R_i = g^{r_i}$ and $h_{i1} = H_3(TS_i \parallel PID_i \parallel y)$, $h_{i2} = H_6(w \parallel n_{t,i} \parallel D_i \parallel Q_i \parallel R_i)$, $\tau_i = r_i + h_{i1}k_i + h_{i2}z_i$. Finally, P_i outputs the signature $\delta_i = (R_i, \tau_i)$ and uploads $\{PID_i, TS_i, \hat{w}_t^{(i)}, w, n_{t,i}, \delta_i\}$ to AS.
- 3) *Verification:* Once receiving the above tuples, AS first checks whether TS_i is fresh. If so, AS continues to verify the validity of the signature. Otherwise, it indicates $\hat{w}_t^{(i)}$ may be from the previous rounds.

Next, AS calculates $h_{i1} = H_3(TS_i \parallel PID_i \parallel y)$ and $h_{i2} = H_6(w \parallel n_{t,i} \parallel D_i \parallel Q_i \parallel R_i)$, then checks if the following equation holds or not.

$$g^{\tau_i} \stackrel{?}{=} R_i(K_i)^{h_{i1}} Q_i^{h_{i2}} \quad (4)$$

If all tuples pass the verification, AS will aggregate the noised model parameters; otherwise, this round of aggregation fails.

- 4) *Aggregation:* AS computes $n_t = \sum_{i=1}^{N\rho} n_{t,i}$, which denotes the total number of samples used in the round. Afterwards, AS aggregates the received local model parameters $\hat{w}_t^{(i)}$ to get the masked global model update

$$\hat{w}_t = \sum_{i=1}^{N\rho} \frac{n_{t,i}}{n_t} \hat{w}_t^{(i)}$$

Finally, AS distributes $\hat{w}_t$ and n_t to all clients.

- 5) *Denoising:* Each client eliminates the mask in $\hat{w}_t$ to get the real global model update

$$\bar{w}_t = \hat{w}_t - \frac{1}{n_t} sum_t.$$

After that, they move on to the next round until the global model converges.

Consider the case that some training users drop out because of the unstable network. Without loss of generality, suppose d clients drop out after receiving the noise vectors. AS will randomly select P_j from the remaining online clients in this round, and transform original ciphertexts of the relevant d noise vectors with $RK_{NPS \rightarrow P_j}$. Next, AS sends these ciphertexts to P_j and waits for the new masked model $\hat{w}_t^{(j)} = w_t^{(j)} + \frac{1}{n_{t,j}} (\varepsilon_{t,j} + \sum_{i=1}^d \varepsilon_{t,i})$. In this case, the overall training instance will be $n'_t = \sum_{i=1}^{N\rho-d} n_{t,i}$. Then AS handles the collected model parameters as usual

$$\hat{w}_t = \sum_{i=1}^{N\rho-d} \frac{n_{t,i}}{n'_t} \hat{w}_t^{(i)}$$

Correspondingly, clients subtract the aggregated noise

$$\bar{w}_t = \hat{w}_t - \frac{1}{n'_t} sum_t$$

VI. CORRECTNESS AND SECURITY ANALYSIS

In this section, we first prove the correctness from three aspects: decryption, signature verification, and model aggregation. Then we analyze the security properties of the privacy-preserving federated learning scheme.

A. Correctness Analysis

1) *The Correctness of Re-Encrypted Ciphertexts Decryption:* The random noise vectors are initially generated and encrypted by NPS. Later, AS converts the ciphertexts so that they can be decrypted by the corresponding clients. Here we demonstrate the correctness of the level-2 decryption.

Given the ciphertext $C' = (C'_1, C'_2, C'_3)$ and the secret key $SK_i = (k_i, z_i)$, P_i first calculates the intermediate parameters

$$\begin{aligned} u'_i &= H_3(ID_\xi \parallel PID_i \parallel (Q_\xi)^{z_i+k_i}) \\ &= H_3(ID_\xi \parallel PID_i \parallel (g^{z_\xi})^{z_i+k_i}) \\ &= H_3(ID_\xi \parallel PID_i \parallel (g^{z_i} g^{k_i})^{z_\xi}) \\ &= H_3(ID_\xi \parallel PID_i \parallel (Q_i K_i)^{z_\xi}) \\ &= u_i \end{aligned}$$

and

$$U = (C'_3)^{u'_i} = (C_3^{RK_{NPS \rightarrow P_i}})^{u_i} = (C_3)^{k_\xi + z_\xi}$$

If the equation $C'_1 = C_1 = g^\alpha = g^{H_2(m \parallel \sigma \parallel ID_\xi \parallel Q_\xi)}$ holds, m can be recovered as follows:

$$\begin{aligned} C'_2 \oplus H_4(U) &= C_2 \oplus H_4((C_3)^{z_\xi + k_\xi}) \\ &= C_2 \oplus H_4((g^\beta)^{z_\xi + k_\xi}) \\ &= C_2 \oplus H_4((Q_\xi \cdot K_\xi)^\beta) \\ &= m \parallel \sigma \end{aligned}$$

Therefore, P_i can decrypt the corresponding transformed ciphertexts correctly and decode them to $\varepsilon_{t,i}$ and sum_t .

2) The Correctness of Client Authentication: Given the tuple $\{PID_i, TS_i, \hat{w}_t^{(i)}, w, \delta_i\}$, AS extracts $PK_i = (D_i, K_i, Q_i)$ to check the client's veracity and message's integrity. The correctness of client authentication can be proved as follows:

$$\begin{aligned} g^{\tau_i} &= g^{r_i + h_{i1} k_i + h_{i2} z_i} \\ &= R_i(g^{k_i})^{h_{i1}} Q_i^{h_{i2}} \\ &= R_i(K_i)^{h_{i1}} Q_i^{h_{i2}} \end{aligned}$$

If the tuple has not been modified during the transmission, it will pass the verification, which indicates that the selected client P_i has submitted the local model successfully.

3) The Correctness of Lossless Aggregation: In the secure aggregation phase, the individual local model parameters are noise-added and aggregated. Then clients download the global model and remove the noise locally. The correctness of lossless aggregation is shown as follows:

$$\begin{aligned} \bar{w}_t &= \hat{w}_t - \frac{1}{n_t} sum_t \\ &= \left(\sum_{i=1}^{N\rho} \frac{n_{t,i}}{n_t} \hat{w}_t^{(i)} \right) - \frac{1}{n_t} sum_t \\ &= \left(\sum_{i=1}^{N\rho} \frac{n_{t,i}}{n_t} \left(w_t^{(i)} + \frac{1}{n_{t,i}} \varepsilon_{t,i} \right) \right) - \frac{1}{n_t} sum_t \\ &= \left(\sum_{i=1}^{N\rho} \frac{n_{t,i}}{n_t} w_t^{(i)} \right) + \frac{1}{n_t} \sum_{i=1}^{N\rho} \varepsilon_{t,i} - \frac{1}{n_t} sum_t \\ &= \sum_{i=1}^{N\rho} \frac{n_{t,i}}{n_t} w_t^{(i)} \end{aligned}$$

In the case of d clients dropping out,

$$\begin{aligned} \bar{w}_t &= \hat{w}_t - \frac{1}{n'_t} sum_t \\ &= \left(\sum_{i=1}^{N\rho-d} \frac{n_{t,i}}{n'_t} \hat{w}_t^{(i)} \right) - \frac{1}{n'_t} sum_t \\ &= \frac{n_{t,j}}{n'_t} \hat{w}_t^{(j)} + \left(\sum_{i=1, i \neq j}^{N\rho-d} \frac{n_{t,i}}{n'_t} \hat{w}_t^{(i)} \right) - \frac{1}{n'_t} sum_t \\ &= \frac{n_{t,j}}{n'_t} \left(w_t^{(j)} + \frac{1}{n_{t,j}} \left(\varepsilon_{t,j} + \sum_{i=1}^d \varepsilon_{t,i} \right) \right) \\ &\quad + \left(\sum_{i=1, i \neq j}^{N\rho-d} \frac{n_{t,i}}{n'_t} \left(w_t^{(i)} + \frac{1}{n_{t,i}} \varepsilon_{t,i} \right) \right) - \frac{1}{n'_t} sum_t \\ &= \sum_{i=1}^{N\rho-d} \frac{n_{t,i}}{n'_t} w_t^{(i)} + \frac{1}{n'_t} sum_t - \frac{1}{n'_t} sum_t \\ &= \sum_{i=1}^{N\rho-d} \frac{n_{t,i}}{n'_t} w_t^{(i)} \end{aligned}$$

Whether there are dropouts or not, clients can accurately eliminate the sum of the noise in the global model. Therefore, the introduction of blind masking will not affect the global model accuracy.

B. Security Analysis

In this part, the security of the scheme is analyzed in detail.

1) **Model Confidentiality:** The model privacy is well protected through the random noise and proxy re-encryption mechanism. To be precise, the veritable local model parameter $w_t^{(i)}$ will not be exposed to any other parties, and the global model update $\bar{w}_t$ is only visible to clients.

Semi-trusted AS cannot infer $w_t^{(i)}$ from $\hat{w}_t^{(i)}$, nor $\bar{w}_t$ from $\hat{w}_t$. First, although AS holds both original ciphertexts and transformed ciphertexts of the noise vectors, it fails to decrypt them without the private keys of NPS and P_i . Therefore, when AS receives the masked local model parameter $\hat{w}_t^{(i)}$, it cannot derive the accurate local model parameter $w_t^{(i)}$. Similarly, since AS is unable to decrypt the ciphertext of sum_t , the global update $\bar{w}_t$ will not be learned.

Any legal clients cannot independently acquire $w_t^{(i)}$, except P_i . If $P_i (i \in S)$ drops out after receiving $C_{\varepsilon_{t,i}}$, AS needs to re-select another online client $P_j (j \in S, j \neq i)$ and securely shares $\varepsilon_{t,i}$ with P_j . In this way, P_j can decrypt the noise vector that should have been assigned to P_i . However, P_i will not upload the masked local model, then P_j has no access to $w_t^{(i)}$.

An external adversary $\mathcal{A}$ cannot obtain $w_t^{(i)}$ and $\bar{w}_t$, even if it captures the data transmitted between entities.

TABLE II
FUNCTIONALITY COMPARISON

Scheme	Confidentiality	Unforgeability	Non-interactivity	Accuracy	Robustness	Server setting
FedOA [30]	✓	×	✓	×	×	Single-Server
Yin [31]	✓	×	✓	×	×	Single-Server
SAFElearn [32]	✓	×	×	✓	✓	Dual-Server
GALAXY [33]	✓	×	×	✓	✓	None
Ours	✓	✓	✓	✓	✓	Dual-Server

Because $\mathcal{A}$ cannot decrypt these ciphertext data and eliminate noise perturbation without the private key, $w_t^{(i)}$ and $\bar{w}_t$ are invisible to $\mathcal{A}$.

- 2) *Conditional Anonymity*: Each client's true identity is concealed behind the changeable pseudonym. Given a disputable client, only the faithful TRA is capable of tracing the real identity

$$ID_i = PID_{i,2} \oplus H_0((PID_{i,1})^\mu || TS_i).$$

Owing to the hardness of DLP, no adversary can figure out μ from K_{pub} so that the real identity will not be exposed. Thus, the scheme provides conditional anonymity to guarantee clients' identities.

- 3) *Resistance of MITM Attack*: In our scheme, the resistance of MITM attacks is guaranteed by the secure CLS, which can withstand the public key replacement attacks and the partially credible KGC. (The security analysis of CLS is similar to the scheme [37], and we do not prove it detailedly here.) Specifically, because CLS enables AS to check the provenance and integrity of the received models, the MITM attackers cannot modify or falsify model parameters successfully. For instance, if an adversary $\mathcal{A}$ intercepts a legitimate client's local model and attempts to alter it to $[w_t^{(i)}]$, AS will discover the signature of $[w_t^{(i)}]$ is invalid by verifying equation (4) and reject the counterfeit model parameters.

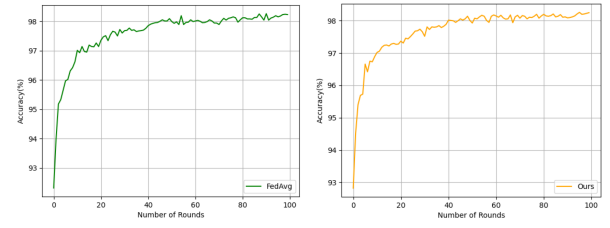
VII. PERFORMANCE ANALYSIS

In this part, we first demonstrate the functionality of the proposed scheme with several existing privacy-preserving federated learning schemes. Additionally, we compare the computational complexity of the selected schemes. In the end, the communication overhead in different phases is evaluated.

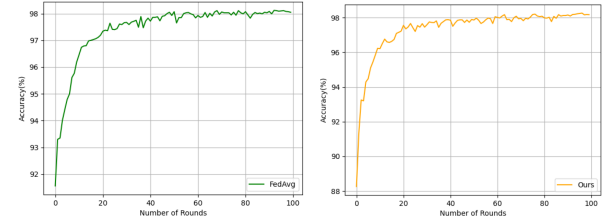
A. Comparison of Functional Features

To clearly illustrate the functional advantages of the proposed scheme, we make a detailed comparison with four existing works [38], [39], [40], [41]. Table II shows the feature comparison involving data confidentiality, unforgeability, non-interactivity, model accuracy, robustness, and server setting.

As shown in Table II, all five works can deliver model confidentiality. FedOA [38] focuses on personalized privacy-preservation of federated learning, in which clients can select their privacy budget and add customized Laplacian noise to the local gradients. Although FedOA has optimized the aggregation results through unbiased estimation of the global gradient, it cannot entirely eliminate the negative impact of noise. Yin [39]



(a) Normal



(b) Dropouts

Fig. 4. Comparison of model accuracy.

introduces multi-input function encryption and Bayesian differential privacy to provide a hybrid privacy protection mechanism. However, the above schemes do not take user dropouts into account. SAFElearn [40] is a generic federated learning architecture instantiated by SMPC. It requires two servers to achieve secure aggregation and dropouts tolerance, which incurs two extra rounds of interaction besides exchanging shares in each training iteration. GALAXY [41] employs threshold secret sharing to support privacy preservation and fault tolerance. Although GALAXY removes the reliance on the server, it still demands multiple rounds of interaction between clients for secret reconstruction.

In contrast to these works, our scheme neither requires multiple interactions nor sacrifices the model accuracy. To demonstrate more intuitively that the proposed scheme is lossless, we conduct simulations related to federated learning. Fig. 4(a) shows the global model accuracy in the normal case, while Fig. 4(b) shows the accuracy in the case of dropouts. As shown in Fig. 4, in contrast to directly transmitting the model parameters, the blind-masking mechanism in both cases does not affect the model accuracy. Meanwhile, the proposed scheme provides client condition anonymous authentication to protect identity privacy, which also supports the unforgeability of the model. Thereby, our scheme provides more comprehensive functionalities for secure federated learning in IoMT.

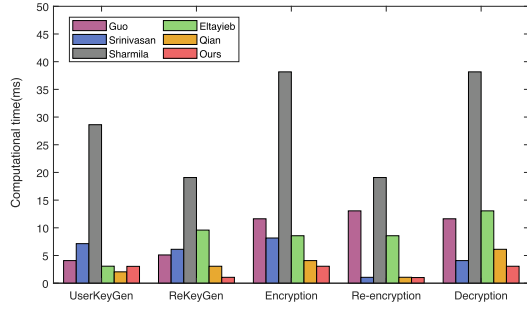
B. Computation Cost

The major computation cost of the scheme comes from proxy re-encryption, so we focus on the performance analysis of PRE and choose several representative schemes [29], [30], [42], [43], [44] for comparison.

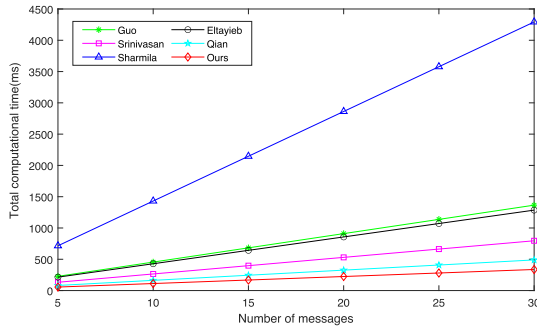
Table III lists the computation cost of the six schemes in the stages of user key generation, re-encryption key generation, encryption, re-encryption, and decryption. Here, t_{exp} denotes a modular exponential operation on multiplicative cyclic group, t_{pair} denotes bilinear pairing, and t_{sm} represents scalar multiplication. In addition, we ignore the negligible operations such

TABLE III
COMPUTATION COST COMPARISON OF EACH SCHEME

Scheme	UserKeyGen	ReKeyGen	Encryption	Re-encryption	Decryption	Total	Assumption
Guo [22]	$4t_{exp}$	$5t_{exp}$	$t_{pair} + 5t_{exp}$	$2t_{pair}$	$t_{pair} + 5t_{exp}$	$4t_{pair} + 19t_{exp}$	q-wDBDH
Srinivasan [23]	$7t_{exp}$	$6t_{exp}$	$8t_{exp}$	t_{exp}	$4t_{exp}$	$26t_{exp}$	M-CDH
Sharmila [34]	$3t_{sm}$	$2t_{sm}$	$4t_{sm}$	$2t_{sm}$	$4t_{sm}$	$15t_{sm}$	EC-CDH
Eltayieb [35]	$3t_{exp}$	$t_{pair} + 3t_{exp}$	$t_{pair} + 2t_{exp}$	$t_{pair} + 2t_{exp}$	$2t_{pair}$	$5t_{pair} + 10t_{exp}$	DBDH
Qian [36]	$2t_{exp}$	$3t_{exp}$	$4t_{exp}$	t_{exp}	$6t_{exp}$	$16t_{exp}$	CDH
Ours	$3t_{exp}$	t_{exp}	$3t_{exp}$	t_{exp}	$3t_{exp}$	$11t_{exp}$	CDH



(a) Computation cost in different stages



(b) Total computation cost

Fig. 5. Comparison of computation cost.

as element selection and hash mapping. We also presume that the verification of public key and the integrity check of ciphertext are pre-processing steps just like [30].

To intuitively present the computational efficiency of each scheme, we conduct an experimental simulation on a PC (Intel Xeon CPU E3-1230 V2 @ 3.30 GHz, 16 GB RAM) and use JPBC library to evaluate the running time of these operations. The execution result of modular exponential, bilinear pairing, and scalar multiplication are 1.02 ms, 6.53 ms, 9.54 ms, respectively. We present the computation cost of relevant schemes in Fig. 5. Fig. 5(a) shows the detailed comparison in all phases. Obviously, Sharmila's scheme [42] takes the most time in each stage, which imposes a large computational pressure on all entities. Among the remaining five works, Guo's scheme [29] has a poor performance in the re-encryption phase and causes extra overhead for the proxy. Srinivasan's scheme [30] and Eltayieb's scheme [43] have the highest cost in the *UserKeyGen* and *ReKeyGen*, respectively, which is not suitable for scenarios involving a multitude of users.

Apparently, our CL-PRE algorithm shows satisfactory performance in all phases. Fig. 5(b) also indicates that our scheme

has the lowest overall cost as the number of messages increases. More importantly, since the decryption of transformed ciphertexts is carried out by clients, we are concerned with the computation overhead in the decryption phase. It is noticeable that our scheme is superior for mobile clients' decryption. Therefore, the proposed scheme has high efficiency, and it is more applicable to the federated learning of IoMT.

C. Communication Overhead

The communication overhead between various parties will be evaluated in this subsection. Assume that the size of timestamp and local model parameter is S_T and S_M , respectively.

Firstly, during the entity registration phase, each client is required to send the pseudonym to KGC, which incurs $N(2|\mathbb{G}| + S_T)$ costs for pseudonym transmission. Then KGC generates the partial public-private key pair for clients and NPS. The return of each partial key pair costs $3|\mathbb{G}|$. When users receive the partial keys, they set up to independently generate their full key pairs. Here, we ignore the publishing cost of full public keys. Secondly, in the pre-computation phase, NPS transfers the ciphertexts $\{\{C_{e_{t,i}}\}_{i=1}^{N\rho}, C_{sum_t}\}$ together with $\{PID_i, RK_{NPS \rightarrow P_i}\}_{i=1}^N$ to AS, causing $(N\rho + 1)(3|\mathbb{G}| + l)$ and $N(3|\mathbb{G}| + S_T)$ costs, respectively. Next, AS transmits $C'_{e_{t,i}}$ to $N\rho$ clients, C'_{sum_t} to N clients, which introduces $(N\rho + N)(2|\mathbb{G}| + l)$ in total. Thirdly, in the secure aggregation phase, each tuple $\{PID_i, TS_i, \hat{w}_t^{(i)}, w, n_{t,i}, \delta_i\}$ will be uploaded to AS. Then AS distributes the aggregated global model to N clients. So the communication cost of aggregation part is $N\rho(5|\mathbb{G}| + 2S_T + S_M + l) + N(S_M + |\mathbb{G}|)$.

Remark that the key of NPS will not be altered throughout T rounds of training. As a consequence, the theoretical communication overhead of the proposed scheme is

$$Cost = 3|\mathbb{G}| + T[(11N + 10N\rho + 3)|\mathbb{G}| + N(\rho + 1)(2S_T + S_M) + (3N\rho + N + 1)l]$$

VIII. CONCLUSION

In response to the privacy leakage issue in IoMT, we put forward a novel privacy-preserving federated learning scheme based on blind noise masking and proxy re-encryption. In our scheme, a semi-trusted third party generates the random vectors, and securely sends them to clients with the help of the aggregation auxiliary server. Clients leverage the allocated noise to mask their local model, and then precisely remove the total noise in the global model. Moreover, a subtle dropout handling mechanism is provided through the flexible re-select

and re-encryption method. Hence, the scheme enables secure, lossless, and robust model aggregation. In addition, the scheme deploys client authentication to prevent uploaded models from being modified, ensuring the correctness of aggregation results. The performance analysis shows that the proposed scheme is especially suitable for the federated training task of mobile IoMT devices, because of the low computational complexity and strong security.

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